REMOVING A CLOSED IRREDUCIBLE SUBSET FROM A CONNECTED SPACE REQUIRES THE AMBIENT SPACE TO BE IRREDUCIBLE

THE STACKS PROJECT AUTHORS

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Lemma 0.1. *Let X be a connected topological space with a finite number of irreducible components $X_1, \ldots, X_n$. If $n > 1$ there is an $1 \leq j \leq n$ such that $X' = \bigcup_{i \neq j} X_i$ is connected.*

Proof. This is a graph theory problem. Let Γ be the graph with vertices $V = \{1, \ldots, n\}$ and an edge between i and j if and only if $X_i \cap X_j$ is nonempty. Connectedness of X means that Γ is connected. Our problem is to find $1 \leq j \leq n$ such that $\Gamma \setminus \{j\}$ is still connected. You can do this by choosing $j, j' \in E$ with maximal distance and then j works (choose a leaf!). Details omitted. □

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